

Revised Generation Interconnection System Impact Study Report

For Queue Project Q18-0964

Springhill 230 kV
April 26, 2020

1. Introduction

This System Impact Study report documents the results of the analyses performed for interconnection request Q18-0964. The Interconnection Customer proposes a 124.2 MW solar photovoltaic facility interconnecting at the Springhill 230 kV substation in the Ohio area. The request was received on March 14, 2018. The study evaluated system intact and single contingency conditions using the regional planning case, including 264 higher-queued active requests and their assigned network upgrades.

2. Summary of Results

The project was found to contribute (DFAX of 5% or greater) to 2 overloaded facilities. The total estimated cost of the network upgrades allocated to this project is \$16,799,000.

3. Network Impacts

3.1 Generator Deliverability and Single Contingency Analysis

Fishkill Plains 500 kV/345 kV Transformer Bank: 126.5% loading (566 MVA) following the outage of the Tidd to Lerdo 500kV line; DFAX = 8.6%.

For the loss of Filer to Tidd 500kV L, the Tidd to Pelicano 138kV Line loads to 111.2% of its 244 MVA rating. Project DFAX: 8.3%.

3.2 Multiple Facility Contingency Analysis

No additional overloads were identified for multiple facility contingencies.

3.3 Short Circuit and Stability

Short circuit and stability analyses identified no additional network upgrades.

4. Required Network Upgrades and Cost Estimates

Facility	Required Upgrade	Cost Estimate	Allocation to Project
Fishkill Plains 500 kV/345 kV Transformer	Replace transformer with higher-rated unit	\$27,770,000	59% (\$16,356,000)
Pelicano Substation to Tidd 138kV	Rebuild line with higher-capacity conductor	\$1,932,000	23% (\$443,000)
	Total allocated to project		\$16,799,000

This report is a simulated study produced by the gridconstraint research prototype on a public synthetic-but-realistic PJM-footprint case; it is not a PJM document.